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To cite this article: P Villemoes *et al* 1993 *J. Phys. B: At. Mol. Opt. Phys.* **26** 4289

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J. Phys. B: At. Mol. Opt. Phys. 26 (1993) 4289–4299. Printed in the UK

Isotope shifts and hyperfine structure of $^{134-138}$Ba II by fast ion beam–laser spectroscopy

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Received 16 March 1993, in final form 16 July 1993

Abstract. We have measured the isotope shifts of the $5d \rightarrow 6p$ transitions in $^{134-138}$Ba II, of which the $5d_{3/2} \rightarrow 6p_{1/2}$ transition has not been measured earlier. We have also obtained values with improved accuracy of the magnetic dipole and electric quadrupole hyperfine coupling constants of the $6p\ ^2P^o_{1/2,3/2}$ levels in ^{135,137}Ba II. The measurements are performed in the red and orange parts of the optical spectrum using fast ion beam–laser spectroscopy with an improved procedure which avoids calibration of the ion velocity. A comparison with theory is made.

1. Introduction

In laser-induced fluorescence studies of fast ions with overlapping laser and ion beams the resonances originating from various isotopes are separated in frequency by the *isotope shift (IS) and the differential Doppler shift. The differential Doppler shift is* usually large in comparison with the IS and the calibration of the ion velocity is therefore often, together with the uncertainty in the frequency calibration, the main source of systematic uncertainties in IS measurements. This uncertainty may be eliminated by recording resonance spectra both with parallel and antiparallel ion and laser beams. In this scheme, demonstrated by Borghs *et al* (1981), the first-order Doppler shift differences between different isotopes are cancelled through a simple addition of recorded frequency shifts and the uncertainty of the measured IS due to drifts in the acceleration voltage becomes negligible. The measurements of Borghs *et al* were performed on an ion beam without mass separation. One aim of this work is to extend the method to mass separated ion beams in order to facilitate the interpretation of spectra and to reduce collisional background from abundant isotopes present *in an unseparated ion beam. This is important in measurements of IS in atoms with* many isotopes and large differences in abundance like samarium which we are going to investigate. In order to be able to excite different isotopes in the same laser scan we have constructed a switch for the current through the analysing magnet. When the fluorescence from one isotope has been recorded the next isotope is rapidly switched into the beamline by changing the magnet current during the scan of the laser. The high accuracy obtained in this work is mainly a consequence of the choice of the experimental arrangement and of a careful calibration of the free spectral range (FSR) of the Fabry–Perot interferometer (FPI) used as a frequency marker in the laser scans.

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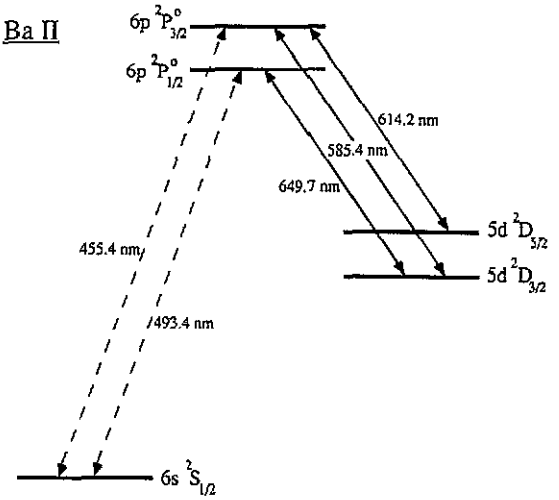


Figure 1. Partial level scheme of Ba II indicating the transitions used (full lines) and discussed (broken lines) in this work.

A suitable element for a test of the technique is Ba ($Z=56$) for which the IS and hyperfine structure (HFS) of transitions between the lowest energy levels in Ba II (see figure 1) have been thoroughly investigated both theoretically (Wilson 1978, Fricke *et al* 1983, Torbohm *et al* 1985, Mårtensson-Pendrill and Ynnerman 1992) and experimentally. The HFS of the ground level 6s $^2S_{1/2}$ and the metastable levels 5d $^2D_{3/2,5/2}$ has been determined very accurately with radio frequency techniques (Werth 1988, Van Hove *et al* 1985, Silverans *et al* 1986, Becker *et al* 1981). The HFS of the 6p $^2P^o_{1/2,3/2}$ levels has been measured by several groups with fast ion beam–laser spectroscopy (Wendt *et al* 1984, Silverans *et al* 1980, Höhle *et al* 1978, Andrä 1976), Lamb-dip spectroscopy (Winter and Galliard 1977) and with Fabry–Perot interferometry (Becker *et al* 1968). Measurements of the IS of the transitions $6s_{1/2} \rightarrow 6s_{1/2,3/2}$ (Wendt *et al* 1984, 1988) and $5d_{3/2,5/2} \rightarrow 6p_{3/2}$ (Van Hove *et al* 1982, Silverans *et al* 1980, Höhle *et al* 1978, Alverez *et al* 1978) have yielded information about the J dependence of the shifts. We have measured for the first time the IS of the $5d_{3/2} \rightarrow 6p_{1/2}$ (649.7 nm) transition and remeasured the IS of the $5d_{3/2,5/2} \rightarrow 6p_{3/2}$ (585.4 nm and 614.2 nm) transitions in the stable isotopes $^{134-138}$Ba II. This enabled us to observe the influence of the 6p $^2P^o_{1/2,3/2}$ levels in the field shift (FS) of the $5d_{3/2} \rightarrow 6p_{1/2,3/2}$ transitions. We have also measured the HFS of the 6p $^2P^o_{1/2,3/2}$ levels and the improved accuracy has established a discrepancy with the theory also for the magnetic dipole constant of the 6p $^2P^o_{3/2}$ level. The measurements have been performed using the proposed improved fast ion beam–laser spectroscopy (FIBLAS) on a beam from an isotope separator.

2. Experiment

Most parts of the experimental set-up have been described elsewhere (Villemoes *et al* 1992) and therefore the description given here is brief. A schematic picture of the apparatus is shown in figure 2. Barium ions, some of which are occupying the metastable levels 5d $^2D_{3/2,5/2}$, are produced in an arc discharge ion source, accelerated to an energy of 35 keV, and isotopically separated by an analysing magnet. The ions

are excited from the metastable levels to the 6p $^2P^0_{1/2,3/2}$ levels by laser light from a CW ring dye laser (CR-699) pumped by an argon-ion laser (INNOVA 400). The resonance is detected by observing the laser induced fluorescence light emitted in transitions from the 6p $^2P^0_{1/2,3/2}$ to the ground level. An elliptical mirror collects the fluorescence and directs it to the entrance aperture of a photomultiplier tube. The elliptical mirror is biased with 300 V defining the interaction region by Doppler tuning the laser light frequency with respect to the accelerated ions. In the case of parallel laser and ion beams the laser light is retroreflected by a mirror.

A portion of the laser light is directed into a temperature stabilized and sealed confocal Fabry–Perot interferometer (FPI) providing frequency markers. The free spectral range (FSR) of the FPI is calibrated against the HFS of the $5d_{3/2} \rightarrow 6p_{1/2}$ transition in ^{135,137}Ba II. In total more than 50 calibration spectra were recorded in the parallel and antiparallel geometries in which the separation between the HFS components $F=2 \rightarrow F'=1$ and $F=1 \rightarrow F'=1$ as well as $F=3 \rightarrow F'=2$ and $F=2 \rightarrow F'=2$ (figure 3) were measured and compared with the high resolution measurements of the HFS of 5d $^2D_{3/2}$ by Van Hove *et al* (1985). The laser scans are always checked for systematic non-linearities by fitting the frequency marker positions to a higher order polynomial. The laser scans were found to be very linear. The result of the FSR is 149.79 ± 0.04 MHz with a 70% confidence interval giving a maximum systematic error of 0.03% to any measured frequency difference in a spectrum. The Doppler shift correction to the separation $\Delta\nu_0$ between HFS components is $\Delta\nu = \pm \Delta\nu_0(v/c) \approx \pm 0.7$ MHz GHz^{-1} (0.07%) for antiparallel (+) and parallel (−) geometry. When recording the same number of spectra in each direction these corrections average out.

The procedure for isotope shift measurements is as follows (figure 4): The laser scan starts and when the first resonance has been recorded (^{134}Ba in the antiparallel geometry) the current through the analyzing magnet is changed with a preadjusted switch to get the next isotope in the beamline (^{135}Ba) and the other isotopes are switched into the beamline subsequently. About ten recordings are made before the

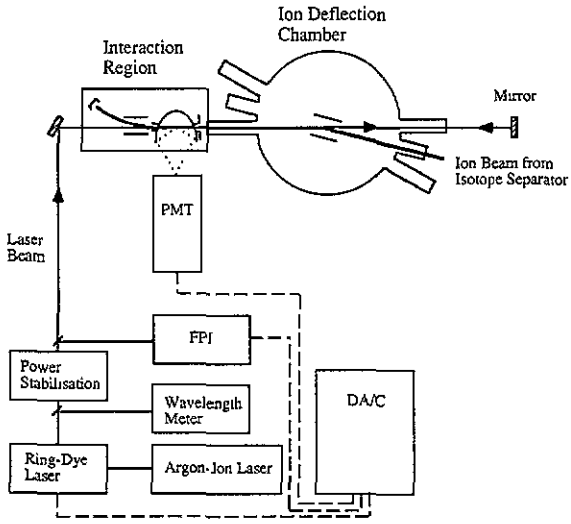


Figure 2. Main features of the experimental set-up for IS and HFS measurements (PMT = photomultiplier tube, FPI = Fabry–Perot interferometer, DA/C = Data acquisition and control).

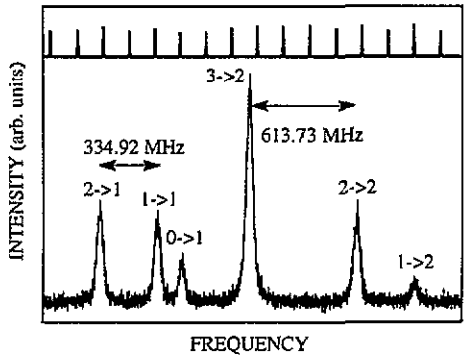


Figure 3. Recorded HFS spectrum for the $5d_{3/2} \rightarrow 6p_{1/2}$ transition in Ba II for the ^{137}Ba isotope used for calibration of the interferometer.

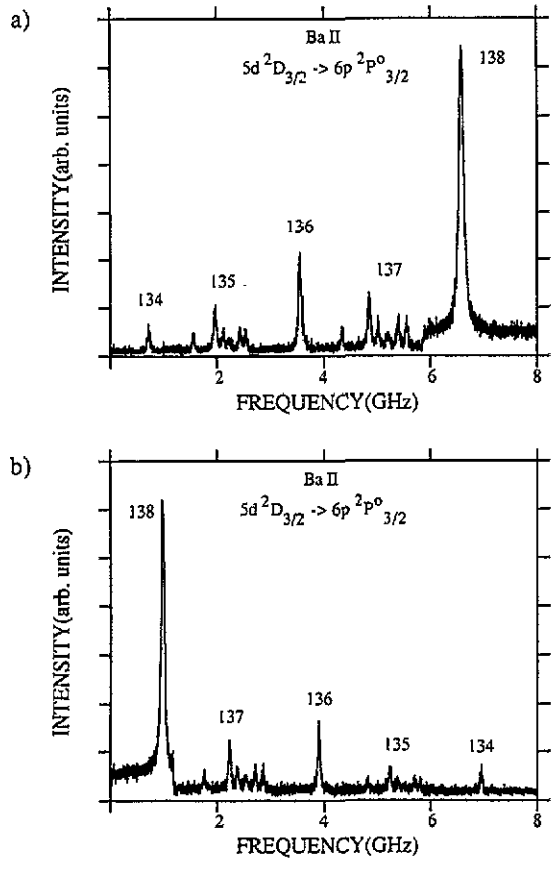


Figure 4. Resonance curves for the isotopes $^{134-138}$Ba II in the transition $5d_{3/2} \rightarrow 6p_{3/2}$. (*a*) Antiparallel laser ion beam interaction. (*b*) Parallel laser ion beam interaction. This set of curves was recorded in one laser scan and with high count rates in order to bring out the low abundance isotopes and to show the advantage of using a mass separated ion beam. Therefore, saturation of the photon counting system causes a seemingly non-natural isotopic abundance and increased linewidth of the ^{138}Ba component.

geometry is changed. The advantage of using a mass separated ion beam instead of a non-separated beam is twofold. Firstly, each part of the spectrum is isotopically pure which simplifies the interpretation if there is an overlap of spectra from different isotopes. Secondly, when the difference in abundance of the isotopes is large the spectrum of an isotope with small abundance might be swamped with the wings of, or with the larger collisionally induced background from, an isotope with large abundance when all isotopes are in the ion beam (the natural abundance of the isotopes investigated here are ^{134}Ba 2.4%, ^{135}Ba 6.6%, ^{136}Ba 7.9%, ^{137}Ba 11.2% and ^{138}Ba 71.7%). As can be seen in figure 4 the height of the peak from ^{134}Ba is about the same as the background from ^{138}Ba. Incidently, valuable information might be lost in case of an overlap of spectra when switching from one isotope to another. This can be avoided by measuring the frequency shift between another pair of isotopes, by measuring the HFS separately or by changing the acceleration voltage.

Spectra in the parallel and antiparallel geometries are recorded within a couple of hours to minimize uncertainties due to experimental conditions such as drifts in acceleration voltage. The relative systematic uncertainty due to drifts in acceleration voltage is given by $\delta v/2c$ where δv is the velocity change caused by the drift and c is the velocity of light. The acceleration voltage provided by our power supply varies less than 0.01% during eight hours which gives a negligible contribution to the maximum systematic uncertainty.

3. Results

A frequency scale is obtained after fitting the position of the frequency markers from the FPI to a polynomial to check for nonlinearities. The positions of the spectral lines are then determined by a curve fitting procedure. The full width at half maximum (FWHM) is typically abut 50 MHz. The HFS of the excited levels could be determined by measuring exclusively the frequency separation between well separated peaks without blending of other HFS components since the HFS of the lower levels is well known. The experimental isotope shift is given by (Borghs *et al* 1981):

$$\delta\nu_{IS} = (\nu_{A'} - \nu_A) = \tfrac{1}{2}(\delta\nu_{AA'}^{\text{parallel}} + \delta\nu_{AA'}^{\text{antiparallel}}) - Q_{AA'} \qquad (1)$$

where $\delta\nu_{AA'}$ is the frequency separation $\nu_{A'} - \nu_A$ between the reference isotope A′ (138 in this work) and the isotope A (134–137) for parallel and antiparallel geometry respectively and $Q_{AA'}$ is a correction for the second-order differential Doppler shift:

$$Q_{AA'} = (\delta\nu_{AA'}^{\text{parallel}} - \delta\nu_{AA'}^{\text{antiparallel}})^2 (M_{A'}^{-1/2} + M_A^{-1/2})/[(M_{A'}^{-1/2} - M_A^{-1/2})8\nu_{A'}]. \qquad (2)$$

The atomic masses $M_{A,A'}$ are from Wapstra and Bos (1977). The value of $Q_{AA'}$ varies between 1 and 4 MHz with an uncertainty of less than 0.1%. The results for the HFS and IS are given in tables 1 and 2 together with previously published data. The uncertainties given within parentheses for our measurements corrspond to a 70% confidence interval and are the linear sum of the estimated maximum systematic uncertainties from the FSR of the FPI and the statistical spread. The IS of the odd isotopes include maximum systematic uncertainties originating from the total uncertainties in the HFS as well. The uncertainties for the previously published data are quoted from the references. In most cases the HFS measurements only include statistical spread (confidence interval not given) whereas for the IS measurements there are often estimates of maximum systematic uncertainties mainly due to high

Table 1. HFS constants for Ba II. The uncertainties presented within parentheses for our measurements include both statistical spread, with a 70% confidence interval, and the estimated maximum systematic uncertainty. The measurements are compared with earlier measurements for which the uncertainties are quoted from the references. The theoretical values are from *ab initio* calculations by Mårtensson-Pendrill and Ynnerman (1992). The values for ^{137}Ba are derived using their values for ^{135}Ba and $g_I^{137}/g_I^{135} = 1.1186$ from Fuller (1976).

Isotope	Level	A (MHz)	B (MHz)	Reference
135	$6p\ ^2P^o_{1/2}$	664.6(0.3)		This work
		665.0(1.6)		Wendt *et al* (1984)
		663.8(2.4)		Becker *et al* (1968)
		621		Theory
	$6p\ ^2P^o_{3/2}$	113.0(0.1)	59.0(0.1)	This work
		113.2(0.8)	59.4(2.3)	Wendt *et al* (1984)
		113.0(1.3)	55.3(3.4)	Silverans *et al* (1980)
		112.7(0.8)	62.1(1.2)	Höhle *et al* (1978)
		113.1(0.5)	60.7(1.2)	Winter and Galliard (1977)
		112.7(0.3)	60.4(0.4)	Andrä (1976)
		113.0(2.1)	63.0(9.0)	Becker *et al* (1968)
		112.4		Theory
137	$6p\ ^2P^o_{1/2}$	743.7(0.3)		This work
		744.1(1.6)		Wendt *et al* (1984)
		741.9(1.3)		Becker *et al* (1968)
		695		Theory
	$6p\ ^2P^o_{3/2}$	127.2(0.2)	92.5(0.2)	This work
		126.7(1.1)	95.0(3.7)	Wendt *et al* (1984)
		127.1(0.6)	89.7(1.5)	Silverans *et al* (1980)
		125.8(0.5)	91.6(1.2)	Winter and Galliard (1977)
		126.0(0.4)	92.5(0.4)	Andrä (1976)
		124.8(0.7)	97.1(9.3)	Becker *et al* (1968)
		125.7		Theory

voltage calibration. Studying the A factors for the ^{135}Ba $6p\ ^2P^o_{3/2}$ level we observe that while previous experimental results were consistent, within error bars, with theory, our more accurate result establish a small discrepancy between experiment and theory. From our IS measurements of the $5d_{3/2} \rightarrow 6p_{1/2,3/2}$ transitions we can derive the difference in IS between the levels $6p\ ^2P^o_{1/2}$ and $6p\ ^2P^o_{3/2}$, e.g. for the isotope pair 138–134 we get 13.6 (1.0) MHz consistent with 11.3 (4.8) MHz derived from Wendt *et al* (1984) in the $6s_{1/2} \rightarrow 6p_{1/2,3/2}$ transitions.

4. Interpretation of results

The contributions to the isotope shift are the mass shift (MS) and the field shift (FS) where the MS is defined to be the sum of the normal mass shift (NMS) and the specific mass shift (SMS) (Bauche and Champeau 1976, Aufmuth *et al* 1987). With the NMS subtracted from the IS we get the residual isotope shift $\delta\sigma$ which can be written as:

$$\delta\sigma = K(M_{A'} - M_A)/(M_{A'}M_A) + F\lambda^{AA'} \tag{3}$$

The first term corresponds to the SMS where K is a purely electronic factor depending on the wavefunctions of the electronic states of the transition. The second term corresponds to the FS where F is also a purely electronic factor proportional to the difference between the electronic densities at the nucleus for the two levels of the transition whereas $\lambda^{AA'}$ is a nuclear factor in first order proportional to the difference $\delta\langle r^{-2}\rangle^{AA'}$ of the mean square radii of the two isotopes. In order to separate the field and mass contributions to the isotope shift a King plot (King 1963) can be made. In a King plot, the modified residual IS ($\delta\sigma' = 2AA'\delta\sigma/(136\times 138(A'-A))$) for a transition α is plotted against $\delta\sigma'$ for another transition β for all pairs AA'. The points

Table 2. Experimental IS for transitions in Ba II. The uncertainties presented within parentheses for our measurements include both statistical spread with a 70% confidence interval and the estimated maximum systematic uncertainty. The measurements are compared with earlier measurements for which the uncertainties are quoted from the references.

Lower level	Upper level	λ_{vac} (nm)	Isotope A	$\delta\nu^{A,138}$ (MHz)	Reference
$5d\ ^2D_{3/2}$	$6p\ ^2P^o_{1/2}$	649.7	134	−174.5(0.8)	This work
			135	−82.7(0.6)	This work
			136	−68.0(0.5)	This work
			137	13.0(0.4)	This work
$5d\ ^2D_{5/2}$	$6p\ ^2P^o_{3/2}$	614.2	134	−192.1(0.7)	This work
				−194.7(0.3)	Van Hove *et al* (1982)
				−212(12)	Silverans *et al* (1980)
				−192(7)	Alvarez *et al* (1978)
			135	−101.4(0.8)	This work
				−103.8(0.3)	Van Hove *et al* (1982)
				−111(11)	Silverans *et al* (1980)
			136	−78.7(0.6)	This work
				−80.3(0.3)	Van Hove *et al* (1982)
				−81(10)	Silverans *et al* (1980)
				−94(4)	Alvarez *et al* (1978)
			137	2.3(0.4)	This work
				3.2(0.3)	Van Hove *et al* (1982)
				0(7.5)	Silverans *et al* (1980)
$5d\ ^2D_{3/2}$	$6p\ ^2P^o_{3/2}$	585.4	134	−188.1(0.6)	This work
				−188.2(0.3)	Van Hove *et al* (1982)
				−209(10)	Silverans *et al* (1980)
				−141	Höhle *et al* (1978)
				−195(9)	Alvarez *et al* (1978)
			135	−97.0(0.7)	This work
				−96.9(0.3)	Van Hove *et al* (1982)
				−115(9)	Silverans *et al* (1980)
				−65	Höhle *et al* (1978)
			136	−76.3(0.4)	This work
				−76.1(0.3)	Van Hove *et al* (1982)
				−85(7)	Silverans *et al* (1980)
				−57	Hohle *et al* (1978)
				−101(8)	Alvarez *et al* (1978)
			137	5.3(0.5)	This work
				7.4(0.3)	Van Hove *et al* (1982)
				1(5.5)	Silverans *et al* (1980)
				−14	Höhle *et al* (1978)

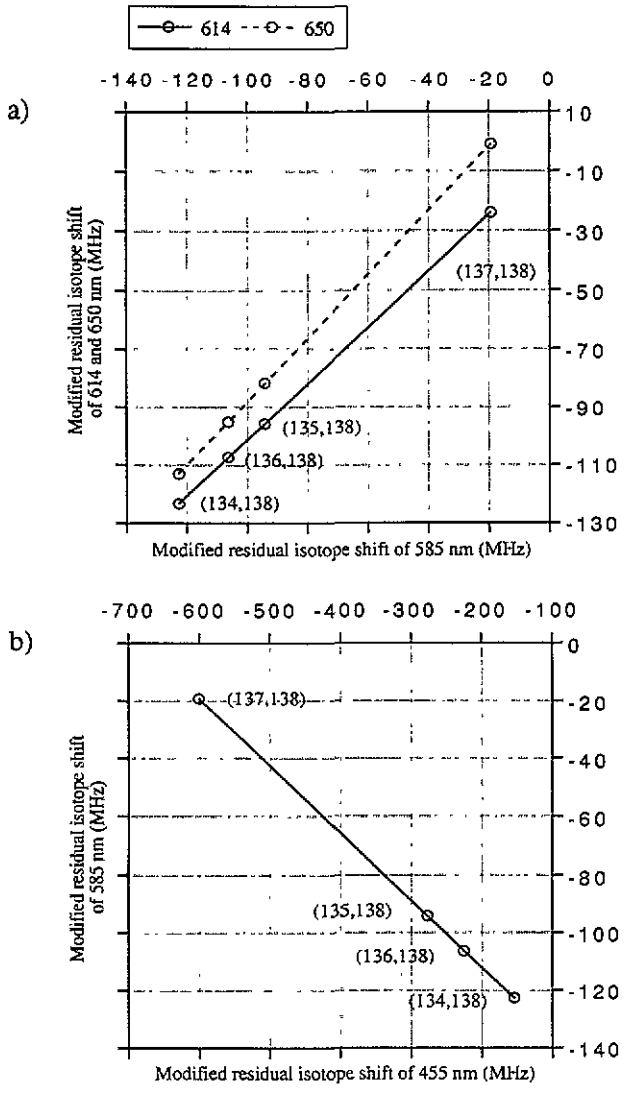


Figure 5. King plots of modified residual is. (*a*) The transitions $5d_{5/2} \rightarrow 6p_{3/2}$ (614 nm) and $5d_{3/2} \rightarrow 6p_{1/2}$ (650 nm) are plotted against the transition $5d_{3/2} \rightarrow 6p_{3/2}$ (585 nm). (*b*) The transition $5d_{3/2} \rightarrow 6p_{3/2}$ (585 nm) against the transition $6s_{1/2} \rightarrow 6p_{3/2}$ (455 nm) measured by Wendt *et al* (1984).

should lie on a straight line according to the formula (^{136}Ba and ^{138}Ba is taken as a standard pair of isotopes):

$$\delta\sigma'_\beta = (F_\beta/F_\alpha)\delta\sigma'_\alpha + (K_\beta - K_\alpha F_\beta/F_\alpha)(M_{138} - M_{136})/(M_{138}M_{136}). \qquad (4)$$

which therefore also provides a consistency check of the measurements. The gradient of the line is the parameter F_β/F_α and from the intersection of the line with the y-axis we obtain the parameter $(K_\beta - K_\alpha K_\beta/F_\alpha)$ (Bauche and Champeau 1976, Aufmuth *et al* 1987). In figure 5(*a*) $\delta\sigma'$ of the transitions $5d_{5/2} \rightarrow 6p_{3/2}$ (614 nm) and $5d_{3/2} \rightarrow 6p_{1/2}$ (650 nm) are plotted against $\delta\sigma'$ of the transition $5d_{3/2} \rightarrow 6p_{3/2}$ (585 nm). The resulting

King parameters from weighted fits are (the uncertainties presented within parentheses are the standard errors from the fits):

$$F_{614}/F_{585} = 0.961(3) \qquad (K_{614} - K_{585}F_{614}/F_{585}) = -49(3)\ \text{GHz amu}$$
$$F_{650}/F_{585} = 1.087(4) \qquad (K_{650} - K_{585}F_{650}/F_{585}) = 191(4)\ \text{GHz amu}.$$

Thus the field shift of the transition $5d_{5/2} \rightarrow 6p_{3/2}$ (614 nm) is 3.9% smaller than that of the transition $5d_{3/2} \rightarrow 6p_{3/2}$ (585 nm). This value compares with 4.4% from Van Hove *et al* (1982). The field shift of the transition $5d_{3/2} \rightarrow 6p_{3/2}$ (585 nm) is 8.0% smaller than that of the transition $5d_{3/2} \rightarrow 6p_{1/2}$ (650 nm).

In order to compare our experimental results with calculations we made a King plot (figure 5(*b*)) of our data for the transition $5d_{3/2} \rightarrow 6p_{3/2}$ (585 nm) against the published experimental values of the IS for the transition $6s_{1/2} \rightarrow 6p_{3/2}$ (455 nm) (Wendt *et al* 1984). It should be mentioned that the IS by Wendt *et al* (1984) are related to the reference isotope ^{138}Ba as $\delta\nu_{IS} = \nu_A - \nu_{138}$ whereas we use $\delta\nu_{IS} = \nu_{138} - \nu_A$ according to Aufmuth *et al* (1987). From the plot we get $F_{585}/F_{455} = -0.2312(6)$ in agreement with $\Delta|\Psi(0)|^2_{5d\rightarrow 6p}/\Delta|\Psi(0)|^2_{6s\rightarrow 6p} = -0.23$ calculated by Wilson (1978) and $(K_{585} - K_{455}F_{585}/F_{455}) = -1483.0(1.4)$ GHz amu. The fact that the electronic part of the field shift is negative and not negligible in the $5d \rightarrow 6p$ transitions in Ba II is interpreted as a screening of the potential from the core s electrons by the 5d electrons and has been discussed earlier (Van Hove *et al* 1982, Silverans *et al* 1980, Höhle *et al* 1978, Wilson 1978). The change in electronic density at the nucleus for the transition $6s_{1/2} \rightarrow 6p_{3/2}$ (455 nm) has been calculated with different results: Torbohm *et al* (1985) obtained the value $F_{455} = -3.83$ GHz fm^{-2} whereas Mårtensson-Pendrill and Ynnerman (1992) obtained $F_{455} = F_{6s_{1/2}} - F_{6p_{3/2}} = -4.13(13)$ GHz fm^{-2} by including some correlation effects, and their value approaches the semi-empirical estimate $F_{455} = -5.1(3)$ GHz fm^{-2} by Wendt *et al* (1988). Using the most recent value of $F_{455} = -4.13(13)$ GHz fm^{-2} by Mårtensson-Pendrill and Ynnerman (1992)† which inludes $F_{6p_{3/2}} = 0.172(1)$ GHz fm^{-2} we can obtain, using the ratios between F values derived from our King plots:

$$F_{585} = 0.95(3)\ \text{GHz fm}^{-2} \qquad F_{614} = 0.92(3)\ \text{GHz fm}^{-2} \qquad F_{650} = 1.04(4)\ \text{GHz fm}^{-2}$$
$$F_{5d_{3/2}} = F_{585} + F_{6p_{3/2}} = 1.13(3)\ \text{GHz fm}^{-2}$$
$$F_{5d_{5/2}} = F_{614} + F_{6p_{3/2}} = 1.09(3)\ \text{GHz fm}^{-2}$$
$$F_{6p_{1/2}} = F_{5d_{3/2}} - F_{650} = 0.09(5)\ \text{GHz fm}^{-2}.$$

The value of $F_{6p_{1/2}} = 0.09(5)$ GHz fm^{-2} compares with the theoretical value 0.111(8) GHz fm^{-2} by Mårtensson-Pendrill and Ynnerman (1992). Calculating the ratio F_{493}/F_{455} from our derived value of $F_{6p_{1/2}}$ and the theoretical values of F_{455} and $F_{6p_{3/2}}$ used above we obtain $F_{493}/F_{455} = 0.98(4)$ which compares well with the experimental ratio of 0.975(3) by Wendt *et al* (1984) and the theoretical value $F_{493}/F_{455} = 0.99(4)$ by Mårtensson-Pendrill and Ynnerman (1992). Theory and experiment for the field shift are seen to be consistent.

The SMS for the $5d \rightarrow 6p$ transition is expectd to be large (Aufmuth *et al* 1987). Wilson (1978) has calculated the SMS for $5d \rightarrow 6p$ to be $-72\,\Delta A$ MHz which gives -144 MHz for the isotope ^{136}Ba with respect to ^{138}Ba, whereas Pendrill (1984) used semi-empirical estimates by a graphical method to obtain $-152(14)$ MHz for the SMS.

† The value of $F_{6p_{3/2}}$ given by Mårtensson-Pendrill and Ynnerman (1992) is too large by a factor $\sqrt{2}$ (Mårtensson-Pendrill 1993).

The SMS for the $6s \rightarrow 6p$ transition is expected to be small (Aufmuth *et al* 1987) so if we assume that it is zero in $(K_{585} - K_{455}F_{585}/F_{455}) = -1483.0(1.4)$ GHz amu we can derive the SMS for the $d \rightarrow p$ transitions, e.g. $\delta\nu_{SMS}$ (585 nm) $= -158$ MHz, which is about -5 times the NMS and consistent with Wilson (1978) and Pendrill (1984). Mårtensson-Pendrill and Ynnerman (1992) have evaluated the electronic part of the SMS for the $6s_{1/2} \rightarrow 6p_{3/2}$ transition using two independent sets of muonic data showing that there are large discrepancies between the data sets. Mårtensson-Pendrill (1993) have also calculated the SMS for all five $6s \rightarrow 6p$ and $5d \rightarrow 6p$ transitions. Using Dirac–Fock (DF) and random phase approximation (RPA) the results are consistent with experiment. However, taking correlations effects into account, the results deviates, e.g. $K_{455} = -115.0$ GHz amu (-12.3 MHz) and $K_{585} = -732.4$ GHz amu (-78.2 MHz) yield $(K_{585} - K_{455}F_{585}/F_{455}) = -759.0$ GHz amu (whereas DF + RPA alone yields -1493.4 GHz amu). Correlation effects are obviously large and difficult to take into account and higher order correlations is probably the cause of this discrepancy with the experimental value.

5. Conclusion

This approach to measure IS combines the advantage of an isotopically pure ion beam with the advantage of combined parallel and antiparallel geometries in FIBLAS. The method yields simple spectra and eliminates the need for an accurate calibration of the ion velocity. This is demonstrated by measuring the IS of the $5d_{3/2} \rightarrow 6p_{3/2}$ (585 nm) and $5d_{5/2} \rightarrow 6p_{3/2}$ (614 nm) transitions. We have also for the first time measured the IS of the $5d_{3/2} \rightarrow 6p_{1/2}$ (650 nm) transition which enabled us to extract the influence of the $6p\ ^2P^o_{1/2,3/2}$ in the FS of the $5d_{3/2} \rightarrow 6p_{1/2,3/2}$ transitions. Our IS measurements of the $5d_{3/2} \rightarrow 6p_{3/2}$ (585 nm) transition agree very well with the measurements by Van Hove *et al* (1982) whereas the IS for the $5d_{5/2} \rightarrow 6p_{3/2}$ (614 nm) transition are about 2% smaller in our measurements. However, it is difficult to compare measurements when confidence intervals are not given and when it is not clear what is included in the estimate of the maximum systematic error. Recent *ab initio* calculations by Mårtensson-Pendrill and Ynnerman (1992) for the field shift are consistent with the experimental results, but the SMS (Mårtensson-Pendrill 1993) seems to be very sensitive to correlation effects.

The HFS measurements of the $6p\ ^2P^o_{1/2,3/2}$ levels in ^{135,137}Ba II are in agreement with earlier measurements and, to our knowledge, the most accurate so far. The result for the $6p\ ^2P^o_{3/2}$ level establishes a small discrepancy with the theory.

Acknowledgments

This work has been supported by the Swedish Natural Science Research Council (NFR) and the Knut and Alice Wallenberg foundation. A-M Mårtensson-Pendrill is acknowledged for providing unpublished isotope shift calculations.

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